TICKET BOOKING PLATFORM

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in
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BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled ”TICKET BOOKING PLATFORM” by S Jishnu Parameswararao (VTU26604) has been carried out in Winter semester of Academic year 2025-2026(WS2526).

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ABSTRACT

The Event Ticket Booking System is a web-based application developed using React JS to simplify and automate the process of booking tickets for internal department events such as seminars, workshops, and technical fests. The system allows users to view event details, enter personal information, and book tickets online through a user-friendly interface. It validates user inputs to ensure accurate booking and prevents overbooking by checking ticket availability. After successful booking, the application displays a booking summary and sends a confirmation email to the user. The system also maintains booking history using local storage, allowing users to view and manage previous bookings. This project reduces manual effort, improves efficiency, and provides a better user experience in event ticket management.

Keywords:

React JS, Event Ticket Booking, Web Application, Online Booking System, Input Validation, Booking History, Email Notification, User Interface.

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LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
CSS	Cascading Style Sheets
DOM	Document Object Model
ES6	ECMAScript 6
HTML	HyperText Transfer Protocol
IDE	Integrated Development Environment
JS	JavaScript
npm	Node Package Manager
UI	User Interface

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Chapter 1

INTRODUCTION

1.1 Introduction

The Event Ticket Booking System is a web - based application it is developed using React JS for managing ticket reservations for internal department events such as seminars, workshops, and technical fests. The main purpose of the system is to provide a simple, fast, and user-friendly platform for users to view event details and book tickets online. In traditional ticket booking methods, the process is mostly manual, time-consuming, and prone to errors such as incorrect data entry and overbooking. This project aims to overcome these issues by automating the booking process and providing instant confirmation to users. The system allows users to enter their details, select the number of tickets required, and complete the booking process efficiently [7]. The application includes important features such as input validation, ticket availability checking, booking summary generation, and book-

ing history management. It also provides user-friendly error messages for invalid inputs and prevents booking more tickets than available. Booking details are stored using browser local storage, allowing users to view and manage their previous bookings. Overall, the Event Ticket Booking System improves efficiency, reduces manual effort, and provides a better user experience for managing event registrations and ticket bookings [8].

1.2 Aim of the Project

The aim of this project is to develop a web-based Event Ticket Booking System using React JS that enables users to view event details and book tickets online in a simple, fast, and efficient manner. The project aims to reduce manual effort, prevent booking errors, validate user inputs, and maintain booking records while providing a user-friendly interface and better booking experience for college events and seminars.

1.3 Scope of the Project

The scope of the Event Ticket Booking System is to provide an efficient and user-friendly platform for managing ticket reservations 2 for internal department events such as seminars, workshops, technical

fests, and cultural programs. The project allows users to view event details, book tickets online, and receive instant booking confirmation. It includes features such as input validation, ticket availability checking, booking summary generation, and booking history management. The system is mainly focused on frontend development using React JS and local storage for data handling. In future, the project can be enhanced by integrating databases, online payment gateways, user authentication, email notifications, and admin management features to support large-scale real-time event management systems.

1.4 Framework

React JS – Frontend JavaScript framework/library used for developing the user interface and building reusable components.Node.js , npm – Used for managing packages and running the React application. HTML5 – Used for creating the structure of web pages.CSS3 – Used for styling and designing the user interface.JavaScript (ES6) – Used for implementing application logic, validations, and interactivity.

Figure 1.1 shows the framework diagram represents the architecture of the Event Ticket Booking System developed using React JS.It shows different modules such as Event Details, Ticket Booking, Booking Summary, and Booking History.The system uses browser local storage

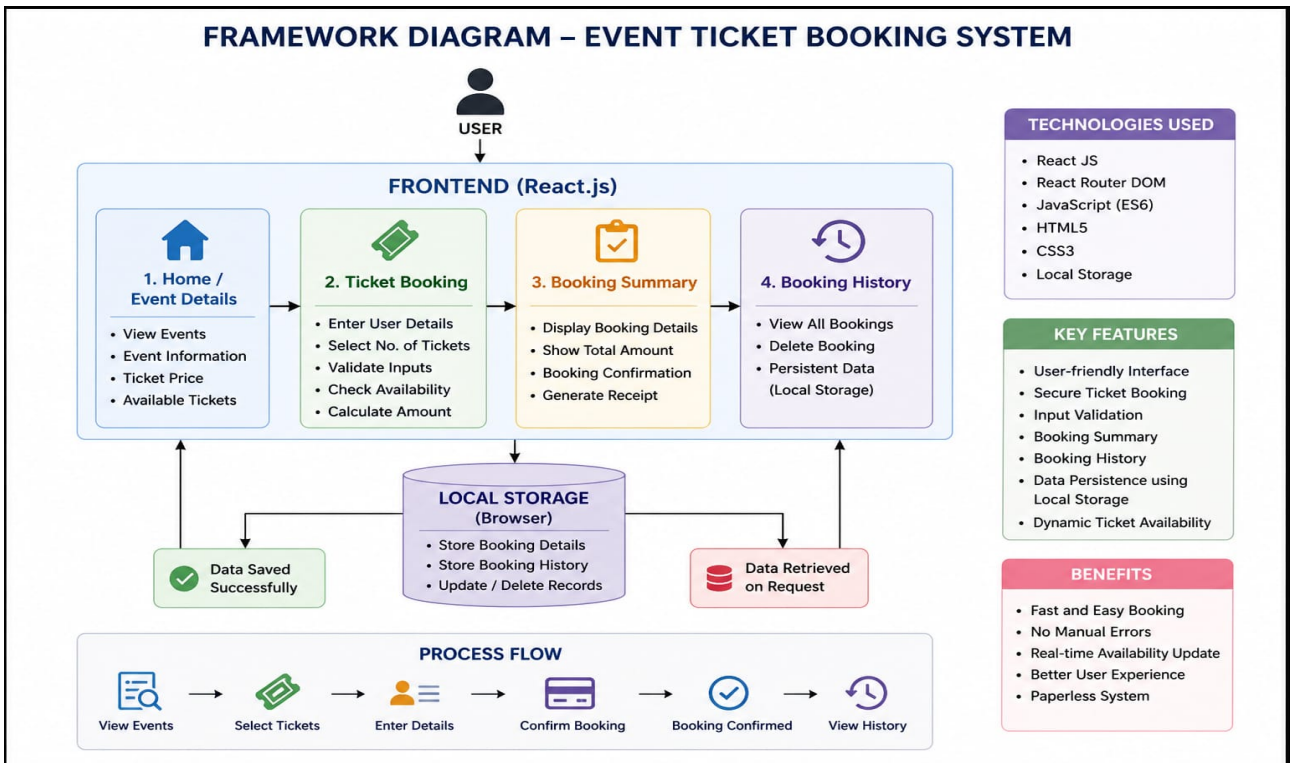

Figure 1.1: Frame work of Ticket booking platform

to save and manage booking information dynamically.Technologies like React JS, JavaScript, HTML5, and CSS3 are used to build the application. The framework provides a user-friendly interface, smooth navigation, and efficient ticket booking management.

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

Many online ticket booking applications are currently available in the market for managing movie, concert, seminar, and event reservations. These applications provide digital ticket booking services with features such as event browsing, seat selection, online payment, and booking confirmation. Some of the popular ticket booking platforms are listed below.

1.BookMyShow :

BookMyShow is one of the most popular online ticket booking platforms in India. It allows users to book tickets for movies, concerts, sports events, and live shows. The platform provides features such as seat selection, online payment, booking history, and e-ticket generation. It is widely used because of its simple interface and large event coverage [1].

2. Eventbrite :

Eventbrite is an international event management and ticket booking platform mainly used for seminars, workshops, conferences, and community events. It allows organizers to create events and users to register online. The platform also provides event analytics, QR-based tickets, and promotional tools [4] .

3. Ticketmaster :

Ticketmaster is a global ticket booking platform used for concerts, sports events, and entertainment shows. It offers online booking, digital tickets, and seat reservation features [3] . The system supports largescale events with real-time booking management and secure ticketing services.

4. Paytm Insider :

Paytm Insider is an Indian event discovery and ticket booking platform that provides access to live events, workshops, comedy shows, and music concerts. It includes online booking, digital payments, and event recommendations [2].

observations from Existing Systems :

Most existing systems provide online booking and digital ticket generation , Many applications support real-time event updates and booking management , Some users report issues such as hidden convenience fees and overpricing , Large-scale systems focus on secure transaction-

sand scalability , Modern ticket booking platforms use QR codes and cloud-based technologies for better performance and security.

Conclusion of Survey :

The survey of existing ticket booking applications shows that online ticket reservation systems are widely used for efficient event management. Inspired by these systems, the proposed Event Ticket Booking System aims to provide a simple, user-friendly, and efficient solution for college events with features such as ticket booking, validation, booking history, and confirmation summary

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend of the Event Ticket Booking System is developed using React JS. The development environment was set up using Node.js and npm (Node Package Manager). Visual Studio Code was used as the code editor for developing and managing the project files. The React application was created using the create-react-app command, which automatically generated the required project structure and dependencies. Google Chrome browser was used for testing and running the application locally. Required packages such as react-router-dom were installed to support page navigation and routing functionality in the application

3.1.2 Structure of the Project

The project follows a component-based architecture using React 7

JS. Different functionalities of the application are divided into separate components and pages to improve maintainability, readability, and code reusability. The project contains components such as Navbar for navigation and pages such as Home, Booking, Summary, and History for implementing different functionalities. The App.js file acts as the main component responsible for routing and managing the application flow, while App.css is used for designing and styling the user interface. The index.js file serves as the entry point of the React application.

3.1.3 Execution Procedure

The execution of the Event Ticket Booking System begins by opening the project folder in Visual Studio Code. The terminal is used to navigate to the project directory and start the React development server using the npm start command. Once the server starts successfully, the application automatically opens in the browser using localhost. The home page displays event details and available tickets. Users can navigate to the booking page, enter their details, and book tickets online. The system validates the entered data, processes the booking, and displays the booking summary and history after successful confirmation.

3.2 Backend Implementation

3.2.1 Environment Setup

The Event Ticket Booking System is mainly developed as a frontendbased application using React JS without a separate backend server. The application uses Node.js and npm only for running and managing the React development environment. Visual Studio Code is used as the code editor for writing and managing the application files. Browser local storage is used to store booking details and booking history instead of using a backend database. Google Chrome browser is used for executing and testing the application locally.

3.2.2 Structure of the Project

Since the project is frontend-based, the application structure mainly consists of React components and pages. Different modules such as Home, Booking, Summary, and History are separated into different files for better organization and maintainability. The application uses React Router for navigation between pages and local storage for temporary data management. Styling and user interface design are handled using CSS files within the frontend project structure.

3.2.3 Execution Procedure

The execution process begins by opening the React project folder in Visual Studio Code. The terminal is used to navigate to the project directory and start the application using the npm start command. Once the React development server starts successfully, the application opens automatically in the browser usinglocalhost. Users can then access

different pages of the application, perform ticket booking operations, view booking summaries, and managebooking history. All booking-related data is processed and stored locally within the browser using local storage.

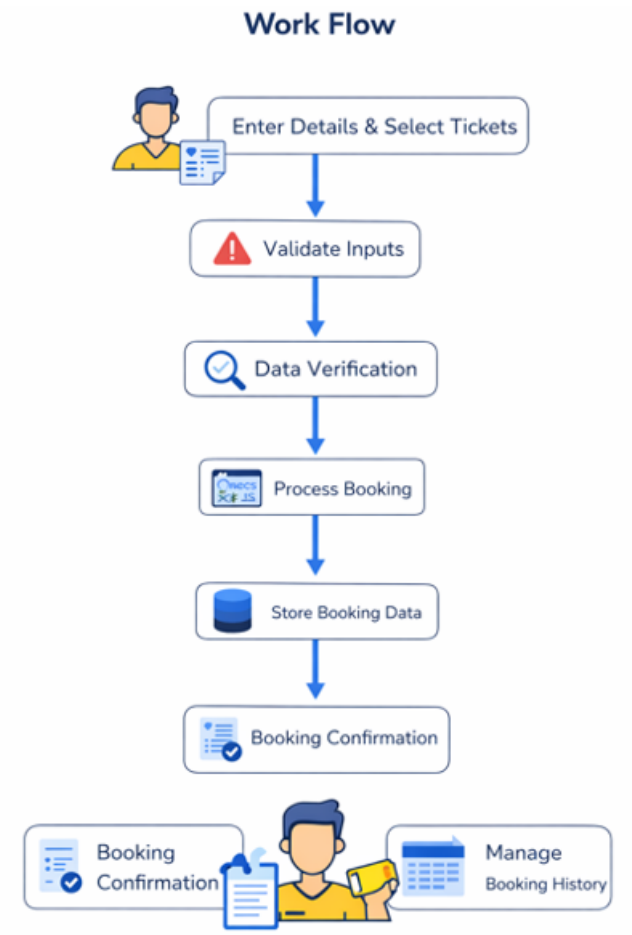

Figure 3.1: Execution procedure of Ticket booking platform

Figure 3.1 shows the workflow diagram illustrates the execution process of the Event Ticket Booking System. It shows the sequence of operations starting from entering booking details and selecting tickets. The system validates user inputs and verifies booking information before processing the request. After successful processing, booking

data is stored in browser local storage and confirmation is displayed. The workflow also allows users to manage and view booking history within the application.

3.3 Database Implementation

3.3.1 Database Configuration The Event Ticket Booking System does not use a separate database. The application uses browser local storage to store booking details and booking history. Local storage is used to save and retrieve data within the browser during application execution.

3.1.2 Schema Structure

The booking data is stored in JSON format inside local storage. The stored data includes user name, email, department, event name, number of tickets booked, total amount, and booking date.

3.1.3 Tables created

No database tables are created in this project because the application is frontend-based and uses local storage instead of a database management system.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The Event Ticket Booking System follows a frontend-based architecture developed using React JS. The application is divided into different modules such as Event Details, Ticket Booking, Booking Summary, and Booking History. Users interact with the application through web pages, and all booking-related data is processed within the React components. Local storage is used to store booking information and retrieve booking history dynamically. The architecture ensures smooth navigation, better user interaction, and modular code organization.

4.2 DataFlow Diagram

The data flow of the system begins when the user accesses the application and views event details. The user enters personal information and ticket details through the booking form. The system validates the 12 entered data and checks ticket availability. After successful validation, the booking is processed and stored in browser local storage. Finally, the application displays booking confirmation, summary details, and booking history to the user.

4.3 Module 1:User Management

This module is responsible for displaying event-related information such as event name, department name, event date, venue, ticket price, and available number of tickets. It provides users with complete details about the event before proceeding with ticket booking.

4.4 Module 2:Event Management

This module handles the ticket booking process by collecting user information such as name, email, department, and number of tickets required. It validates all input fields, checks ticket availability, and prevents invalid bookings or overbooking.

4.5 Module 3:Booking Management

This module displays booking confirmation details including user name, event name, tickets booked, and total amount. It also maintains booking history using local storage and allows users to view or delete specific booking records.

Advantages of this Project:

a) User-Friendly Interface : Provides a simple and interactive interface for easy navigation and booking.

b) Fast and Efficient Booking : The system automates the ticket booking process, reduces manual effort, and provides instant booking confirmation.

c) Validation and Booking Management : The application validates user inputs, prevents overbooking, and maintains booking history for future reference.

Disadvantages:

a) Limited Data Storage : The application uses browser local storage instead of a real database, which limits data handling capabilities.

b) Device Dependency : Booking history and stored data are available only on the same browser and device where the booking was made.

c) Lack of Advanced Security : The system does not include ad-

vanced security features such as user authentication, encryption, or database-level protection.

Chapter 5

RESULTS AND DISCUSSIONS

The Event Ticket Booking System was successfully developed and executed using React JS. The application provides a user-friendly interface for viewing event details, booking tickets, and managing booking history. All modules of the system function correctly and provide smooth navigation between pages [6]. The Event Details module displays information such as event name, department, venue, ticket price, and available tickets. Users can access the booking form through the navigation menu and enter their details for ticket reservation. The Ticket Booking module validates user inputs such as email format and ticket count before processing the booking request. Invalid inputs are handled using appropriate error messages, which improves usability and prevents incorrect bookings. After successful booking, the system displays a booking confirmation summary showing user name, event name, number of tickets booked, and total amount [5]. The Booking History module stores booking details 16 using browser local storage

and allows users to view or delete specific booking records. The application also updates ticket availability dynamically after each booking. The developed system demonstrates the practical implementation of React JS concepts such as functional components, state management using hooks, routing, conditional rendering, and event handling [9]. The project successfully achieves its objective of providing a simple and efficient online ticket booking platform for internal department events.

Table 5.1: System Features Comparison

Feature	**Availability**
Event Details Display	Yes
Online Ticket Booking	Yes
Input Validation	Yes
Email Format Validation	Yes
Booking Confirmation	Yes
Booking Summary	Yes
Booking History	Yes
Delete Booking History	Yes
Responsive User Interface	Yes
Local Storage Support	Yes
Navigation Using React Router	Yes
Dynamic Ticket Availability Update	Yes

Table 5.1 represents the system features implemented in the Event Ticket Booking System. It shows the availability of important functionalities such as event details display, online ticket booking, input validation, and booking confirmation. The system also supports book-

ing history management and deletion of specific booking records. Features like responsive user interface and navigation using React Router improve the overall user experience. The table confirms that all required functionalities are successfully implemented in the application.

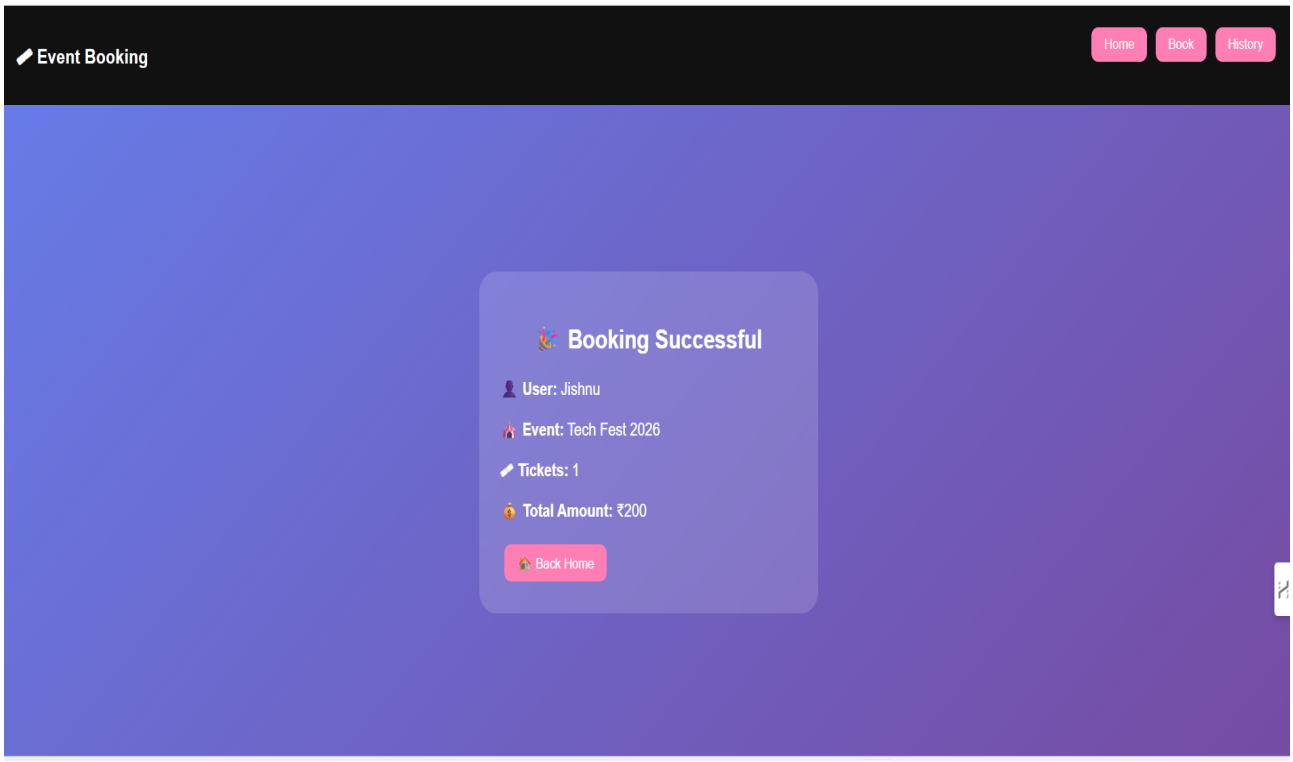

Figure 5.1: Result page of Ticket booking platform

Figure 5.1 shows the booking confirmation page of the Event Ticket Booking System. It displays the user details, event name, number of tickets booked, and total amount after successful booking.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Event Ticket Booking System is a user-friendly web application developed using React JS for managing online ticket reservations for internal department events. The system simplifies the booking process by allowing users to view event details, enter their information, and book tickets efficiently. It validates user inputs, prevents overbooking, and provides instant booking confirmation with booking summary details [10]. The application also maintains booking history using local storage, which improves user convenience and record management. Overall, the project reduces manual effort, improves accuracy, and demonstrates the practical implementation of frontend web development concepts using React JS.

6.2 Future Enhancements

The proposed ticket booking application can be further improved by incorporating advanced features to enhance functionality and user experience. Future enhancements may include the integration of secure online payment gateways to enable seamless transactions, implementation of notification systems such as SMS alerts for booking confirmations and event updates, and development of a dedicated admin dashboard for better event management and analytics. Additional features like QR code-based digital tickets, seat selection visualization, and cancellation/refund options can improve usability. The system can also be extended to support mobile applications, multi-language interfaces, and cloud deployment for better scalability and accessibility. These enhancements will make the application more robust, flexible, and suitable for real-world large-scale usage.

1. BookMyShow :

BookMyShow Official Website

Review Comments: BookMyShow is one of the most popular online ticket booking platforms in India. It provides ticket booking services for movies, concerts, sports events, and live shows. The platform offers a simple user interface, seat selection, online payment, and digital

ticket generation. However, some users report issues related to convenience fees and ticket pricing.

2. Eventbrite :

Eventbrite Official Website

Review Comments: Eventbrite is widely used for seminars, workshops, conferences, and community events. It allows event organizers to create events and manage registrations online. The application provides good event management tools and analytics but may be slightly complex for small-scale users.

3. Ticketmaster:

Ticketmaster Official Website

Review Comments: Ticketmaster is a global ticket booking platform mainly used for concerts and entertainment events. It supports large-scale ticket management and secure online booking. The platform provides real-time updates and digital tickets, but users sometimes experience high service charges and booking traffic issues.

4. Paytm Insider :

Paytm Insider Official Website

Review Comments: Paytm Insider is an Indian event ticket booking platform that offers tickets for music shows, comedy events and live experiences. It provides smooth online booking . The application is user-friendly and well integrated with online payment services.

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Table 7.1: Mapping to Complex Engineering Problems

Level	Attribute	Characteristics	WKs	Justification
WP1	Depth of Knowledge	Requires in-depth engineering	WK4, WK5	Knowledge of React JS, hooks, validation, UI design
WP2	Conflicting Requirements	Technical and non-technical constraints	WK4, WK5	Balances usability with booking and validation
WP3	Depth of Analysis	Analytical and design thinking	WK3, WK4	Component design, state handling, dynamic updates
WP4	Familiarity	Partially new problems	WK4	Real-time booking and ticket updates
WP5	Applicable Codes	Limited standard solutions	WK6	Custom booking and validation logic
WP6	Stakeholder Needs	Multiple stakeholders	WK5, WK6	Students, faculty, organizers
WP7	Interdependence	System-level integration	WK3, WK5	UI + state + validation integration

Table 7.1 shows the mapping of the Event Ticket Booking System to Complex Engineering Problems using React JS concepts such as

state management, validation, and UI design. It demonstrates problem-solving, dynamic booking management, and integration of multiple frontend functionalities.

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

Table 7.2: Mapping to Complex Engineering Activities

EA No.	Attribute	Characteristics	Justification
EA1	Range of Resources	Use of technologies	React JS, HTML, CSS
EA2	Level of Interactions	Complex module interaction	Event display + booking + validation
EA3	Innovation	Modern technologies	Hooks, conditional rendering
EA4	Consequences	Impact on users	Simplifies booking
EA5	Familiarity	Beyond standard	Dynamic real-time system

Table 7.2 describes the mapping of the Event Ticket Booking System to Complex Engineering Activities using modern frontend technologies such as React JS, HTML, and CSS. It highlights component interaction, dynamic booking functionality, validation handling, and user-focused system design.

7.3 SDG Mapping (CEP Section)

Table 7.3: SDG Mapping

SDG No.	SDG Title	Relevance
SDG 4	Quality Education	Supports academic events
SDG 9	Innovation	Promotes React-based systems
SDG 10	Reduced Inequalities	Accessible to all users
SDG 11	Sustainable Communities	Encourages organized events

Table 7.3 describes the mapping of the Event Ticket Booking System to Sustainable Development Goals (SDGs). The project supports quality education, promotes innovative web technologies, provides accessibility for users, and helps in organizing events efficiently within educational communities.

References

[1] BookMyShow Official Website. https://in.bookmyshow.com

[2] Paytm Insider Official Website. https://insider.in

[3] Ticketmaster Official Website. https://www.ticketmaster.com

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[6] SeatGeek Official Website. https://seatgeek.com

[7] TicketLeap Official Website. https://www.ticketleap.com

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[9] Brown Paper Tickets Official Website. https://www.brownpapertickets.com

[10] Eventzilla Official Website. https://www.eventzilla.net